



Smart-contract-based blockchain-enabled decentralized scheme for improving smart-grid security

Bhabendu Kumar Mohanta ^a, Ali Ismail Awad ^{a,*}, Tarek Elsaka ^a,
Hamza Kheddar ^b, Ezedin Baraka ^a

^a College of Information Technology, United Arab Emirates University, Al Ain, P.O. Box 15551, United Arab Emirates

^b LSEA Laboratory, Electrical Engineering Department, Faculty of Technology, University of Medea, 26000, Algeria

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ABSTRACT

Intelligent devices with embedded technology have proliferated dramatically over the past decade. The Internet of Things (IoT) has emerged as a transformational force, advancing traditional systems to previously unattainable levels of intelligence. Smart cities, transportation, healthcare, supply-chain management, agriculture, water management, and smart grid (SG) systems are among the industries where the IoT has found applications. These developments are demonstrated by the integration of IoT systems into SG networks, offering significant improvements in sustainability, dependability, and efficiency. Such systems use various IoT devices to continuously monitor the environment and transmit data for processing and analysis. Nonetheless, the growth of the IoT has introduced security vulnerabilities, including concerns about user identification, data integrity, and trust, especially in SG applications. This study aims to resolve several security challenges in IoT-enabled SG applications to support sustainability. The proposed scheme effectively tackles critical security requirements such as data integrity, user anonymity, distributed storage, trust management, and decentralized architecture. The security concerns addressed by blockchain technology include preserving data integrity, fostering trust, providing secure communication, and enabling effective monitoring. Smart contracts automate system processes and are effective in maintaining user trust. The experimental findings support the viability of the proposed system, demonstrating a computational cost of 3.150 ms and a communication overhead of 992 bits, both representing improvements over various existing solutions. Additionally, the deployment cost for the smart contract is found to be 5.64 USD with a writing cost of 2.89 USD, both of which are lower than the costs associated with comparable approaches.

1. Introduction

Today, economical and reliable electricity has become an essential and unavoidable basic need in a sustainable society. Nonetheless, a notable fraction of the world relies on inefficient conventional electric power distribution systems. Within these systems, power is generated through nuclear, coal, and hydroelectric production units and transmitted through an extensive network of electric power lines [1]. In principle, electrical energy flows in one direction, from the original producers to consumers; however, as the demand

* Corresponding author.

E-mail addresses: bhabendu@uaeu.ac.ae (B.K. Mohanta), ali.awad@uaeu.ac.ae (A.I. Awad), telsaka@uaeu.ac.ae (T. Elsaka), kheddar.hamza@univ-medea.dz (H. Kheddar), ebarka@uaeu.ac.ae (E. Baraka).

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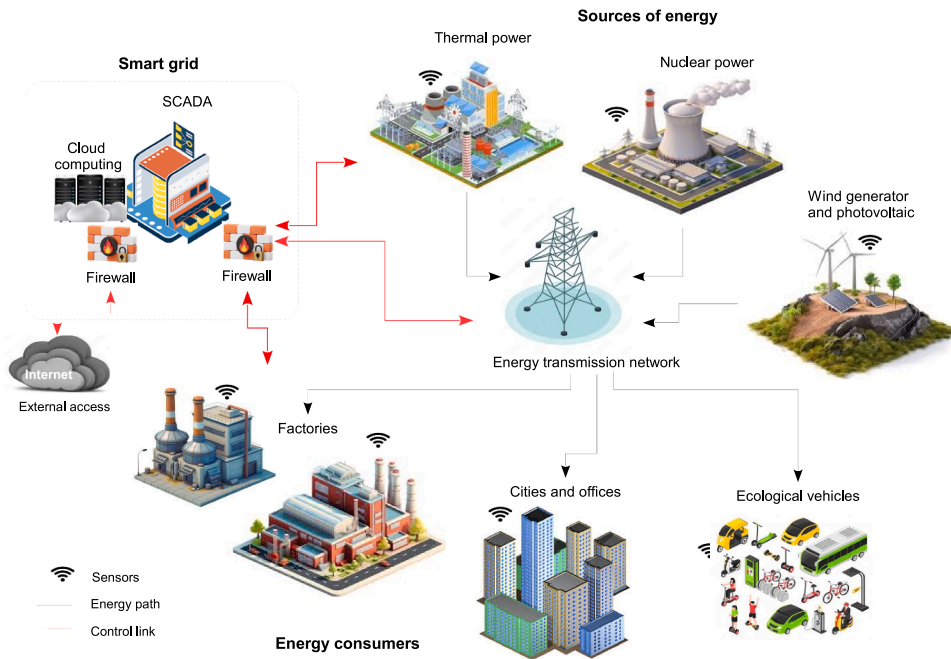


Fig. 1. Block diagram of a smart grid system.

for electricity has grown, traditional power grid systems have become insufficient. This has highlighted the need for more advanced grid systems to address the problem of aging infrastructure, improve reliability, and integrate renewable energy sources.

A smart grid (SG) is a system of interconnected, intelligent devices that enables communication between power-generation systems, distribution infrastructure, and consumers. Smart devices and advanced technologies such as the Internet of Things (IoT), fog computing, and cloud computing can be used to transform a traditional grid network into an SG network. In the current context, a functioning economy and reliable electricity are vital and unavoidable basic needs [2]. Fig. 1 illustrates how energy is produced by sources such as thermal power, nuclear power, and wind generators. Various units—including factories, cities, offices, households, and electric vehicles—both receive and transmit energy. It is evident that with the development of smart devices and innovative technologies, traditional grid networks have begun to evolve into SG systems, incorporating elements such as supervisory control and data acquisition (SCADA). These modern grid systems rely on cloud computing for efficient data storage and processing, enabling real-time monitoring and control. Additionally, renewable energy sources, advanced metering infrastructure, data-acquisition systems, and electric vehicles form integral parts of the SG ecosystem [3]. Furthermore, SG systems offer important advantages for both domestic and industrial electricity users. These advantages include improved efficiency and reliability of electricity supply, and enhanced security in network management and coordination. Ultimately, this leads to better performance and a reduction in carbon emissions [4]. Next-generation energy infrastructure incorporates renewable energy sources, including power generation from photovoltaic cells, fuel cells, wind turbines, and other technologies.

Nonetheless, these architectures face several security challenges, including issues with identification, authentication, data loss, breaches, eavesdropping, denial-of-service (DoS), malware linked to advanced persistent threats (APTs) [5], and other cyber threats. While SGs can offer self-healing capabilities that enhance their resilience, these very features can themselves introduce new vulnerabilities and potential failure points if they are not properly secured. Moreover, because SGs support dynamic supply–demand management [6] and can enable real-time adaptation to fluctuations in energy production and consumption, they rely heavily on continuous data exchange and automated control; this increases the attack surface, meaning that there is an increased need for robust cybersecurity mechanisms. Furthermore, SG technology continues to face critical security challenges, particularly its limited ability to safeguard personal information and ensure the secure transmission and distribution of data, all of which can have direct societal impacts [7]. Personal data is a highly valuable asset, and any leakage can lead to catastrophic consequences for both users and service providers. In response, emerging technologies such as advanced cryptographic protocols and blockchain have been proposed as ways in which to secure data transmission and storage in trusted and verifiable environments, offering potential solutions through robust encryption, consensus mechanisms, and data integrity verification.

Inadequate security can lead to power outages, financial losses, and breaches of data privacy. With the massive flow of data and increasing connectivity of SGs, the risk of cyberattacks, data breaches, and system disruption has grown substantially; this means that the implementation of robust security solutions is not just necessary, but critical. Security measures can be used to protect critical infrastructure from malicious threats and ensure uninterrupted energy delivery, which is vital for economic stability, public safety, and overall societal welfare.

Advanced solutions for mitigating these risks are essential. These include multilayered cybersecurity frameworks, encryption protocols, secure data storage, efficient intrusion-detection mechanisms [8,9], and real-time monitoring tools. Secure SGs also enable rapid responses to threats and natural disasters, helping to maintain the stability of the grid and the availability of energy, even during crises. This is especially important for large cities and megacities, where increasing energy demands require secure and energy-efficient technologies.

1.1. Research motivations

The “Industry 4.0” revolution is transforming modern industries and urban infrastructure through the integration of advanced technologies such as the IoT, artificial intelligence, big data, robotics, and blockchain. These innovations are driving the development of smart cities and SGs; however, ensuring long-term societal sustainability requires not only technological progress but also the establishment of ethical, secure, and socially responsible systems. In this context, blockchain has emerged as a promising solution for enhancing the security and auditability of private data exchange in SG systems [10].

Previous studies in the literature emphasize innovative solutions for ensuring the integrity, security, and privacy of SG environments. These include the use of blockchain technology, cryptographic techniques such as hashing, digital signatures [11], lattice-based encryption [12], verifiable secret sharing [13], secure multiparty computation, elliptic-curve cryptography (ECC) [14], cross-domain data collection schemes such as the verifiable cross-domain data collection and privacy-persevering sharing (VC-DCPS) scheme [15], and ciphertext-policy attribute-based encryption (CP-ABE) [16]. Previous research has highlighted opportunities for secure authentication using anonymous key management and certificateless aggregate signatures [17]. Regarding network architecture, private blockchain has been employed to secure data transfer between IoT-enabled smart meters and service providers [18], providing secure communication, mutual authentication, and tamper-proof data storage, limiting the risks of identity theft and unauthorized access [19]. In IoT-enabled SG systems, blockchain is also crucial for demand response management (as in the blockchain-enabled privacy-preserving scheme; BPPS) and access control [20], providing secure authentication [21] and supporting decentralized operations. Certificate-based access-control mechanisms can also be applied to SG systems [22], improving cyber-physical security in ecosystems with electric vehicles and facilitating smoother grid operations by balancing privacy and openness.

This study sought to investigate the potential of blockchain to establish a secure, intelligent, and resilient energy infrastructure. Blockchain networks leverage consensus protocols and asymmetric cryptography to ensure data confidentiality, transparency, and traceability. Nevertheless, challenges remain, including computational complexity, scalability constraints, system integration, and the high cost of on-chain data storage. To address these limitations, in this paper, we propose a decentralized framework that divides the network into interconnected local and global segments. This framework incorporates edge nodes, smart contracts, and blockchain-enabling technologies to support trusted user registration and improve device authentication and secure communication in edge-computing environments.

The novelty and advantages of the proposed framework over existing approaches are listed in Table 1. In particular, this work distinguishes itself by introducing a blockchain-enabled, edge-integrated framework to specifically address these limitations while enhancing scalability, decentralization, and device-level security in SGs.

1.2. Research contributions

The key contributions of this study can be summarized as follows:

- We present a comprehensive investigation of sustainability and cybersecurity challenges in IoT-enabled SGs, with a focus on integrating edge computing and blockchain to establish a secure and resilient energy infrastructure.
- We propose a novel decentralized architecture featuring hierarchical blockchain layers (local and global), edge nodes for localized processing, and smart contracts to support distributed data management, mutual authentication, and secure peer-to-peer transactions.
- The framework incorporates ECC and zero-knowledge proof techniques to enable lightweight, privacy-preserving authentication and to ensure data integrity in resource-constrained environments.
- The solution is implemented on the Ethereum platform, and empirical evaluations demonstrate improved efficiency and security compared to existing schemes.

1.3. Paper structure

The remainder of this paper is structured as follows. Section 2 delves into previous relevant work in depth. Section 3 discusses the proposed solution. Section 4 presents the outcomes of the proposed solution and an analysis of the results, proposing an enhanced architecture to address the issues. Finally, Section 5 concludes the paper.

2. Related work

Blockchain commoditizes trust, enabling automated smart contracts for auditable multiparty transactions between energy providers and customers. These contracts eliminate the need for third-party intermediaries, facilitating the adoption and monetization of distributed energy and financial transactions. Using blockchain can enhance the fidelity and security of building-to-grid

Table 1

Comparison of various features between our proposed scheme and existing methods in relation to SGs. The symbol ● indicates that a feature is fully supported, while ○ indicates that it is not supported; — indicates that the information is unavailable.

Reference	Edge	DDS	DA	Privacy preservation	Data integrity	Anonymity	Authentication
[12]	●	—	●	●	—	●	●
[15]	●	—	●	●	●	●	●
[17]	—	●	●	●	●	●	●
[18]	—	—	●	—	●	●	●
[20]	—	—	●	●	●	—	●
[21]	●	—	●	—	—	●	●
[24]	—	—	○	●	●	○	●
[25]	—	—	●	—	●	●	●
Proposed	●	●	●	●	●	●	●

Abbreviations: DDS, distributed data store; DA, decentralized architecture.

communications. Additionally, multiple customers can use the blockchain to cryptographically verify each other's data, creating a distributed trust mechanism. Numerous researchers have proposed using original blockchain technology, customized blockchain frameworks, or integrated infrastructures that combine blockchain with other technologies to ensure secure transactions in SGs. A SG is a complex system of systems with intricate interactions among energy, information, and communication subsystems, and smart-metering architecture has the potential to introduce many new security and privacy issues [23].

Islam et al. [24] investigated a security framework for SGs that incorporates advanced cryptography, machine learning (ML), and physical-layer techniques; however, this faces constraints in relation to computational complexity and use of resources, and it was not tested on other layers. Wang et al. [25] proposed a blockchain-based mutual-authentication protocol that sought to address security issues in SGs; this protocol combines blockchain with ECC and dynamic join-and-exit mechanisms. Their scheme was found to improve security and performance through batch verification and Schnorr signatures, but its limitations included its computational complexity and challenges in relation to integration. Mylrea et al. [26] proposed a blockchain-based scheme with smart contracts to secure distributed energy transactions in SG systems, seeking to enhance grid resilience and enabling secure, automated, and auditable energy exchanges without third-party involvement; however, this scheme lacks standardization and adherence to regulatory frameworks, which poses interoperability and security challenges and may reduce customer trust.

Alaguraj and Kathirvel [27] developed an SG system that uses IoT and blockchain technologies to make use of renewable energy sources such as photovoltaics and wind-energy conversion systems. This system uses edge computing for communication and a sharded blockchain architecture for secure and transparent transactions within the SG ecosystem. Wen et al. [28] introduced a blockchain-enhanced price-incentive demand response strategy for sustainable city management; this uses building users as price anticipators and ensures transparency and traceability. Xu et al. [29] developed a secure energy-policy and load-sharing method for renewable alternating current smart microgrids using industrial IoT and cloud-computing technology; they demonstrated the effectiveness of blockchain in enhancing cybersecurity. Kolahan et al. [30] presented a blockchain-based solution for demand-side management of residential buildings; this was found to enhance peak-to-average ratios, reduce energy consumption, and increase thermal comfort. Their solution uses a transient numerical physical model and Ethereum Solidity for the development of smart contracts.

Samuel et al. [31] proposed a secure energy system for vehicles in sustainable cities using private and public blockchains. Their system applies differential privacy in consensus management, reducing energy prices and operating costs, and it provides enhanced privacy protection. Zhu et al. [32] developed a blockchain-based fuzzy sliding-mode controller for off-grid islanding microgrids. This controller reduces steady-state error, improves response time, stabilizes microgrids, and protects networks from attacks, enhancing data security and reducing control chatter. Singh et al. [33] developed a secure blockchain-enabled framework for energy-efficient smart parking in sustainable-city environments. The transport layer employs ECC cryptography for secure communication, while the roadside-unit-based blockchain network ensures distributed authentication.

Wang et al. [36] developed a secure blockchain-based framework for managing renewable hybrid microgrids in multi-agent distributed structures; this enables secure data transactions and reduces operational costs using an optimization algorithm. Deng et al. [34] presented an identity-based designated verifier anonymous aggregate signature (IB-DVAAS) scheme for SGs that enables secure, anonymous data aggregation and verification by a designated data center. This improves on previous approaches by providing security proofs in the standard model; however, it remains a centralized solution that depends on a trusted key generator and may face scalability challenges in practical deployment. In a related effort focused on lightweight cryptographic authentication, Mahmood et al. [35] proposed a lightweight authentication scheme for SG communication based on ECC. This scheme eliminates reliance on a trusted third party (TTP) during authentication; it also supports mutual authentication and ensures user anonymity with low computational overhead. Security is validated using ProVerif and Burrows–Abadi–Needham logic. While this scheme performs efficiently, it is nonetheless centralized due to its dependence on a TTP for initial registration and key distribution; this may limit scalability and dynamic integration in large, decentralized SG environments.

Emerging research in blockchain and SG cybersecurity investigates how distributed ledger technology can foster trust, dependability, and resilience in energy systems. The decentralized structure of blockchain removes single points of failure, and its immutability and consensus processes produce tamper-free and verifiable data from sensors, smart meters, and distributed energy resources (DERs). Multiple studies have examined the potential application of permissioned blockchain frameworks, such as those based on PBFT or

Table 2
Comparison of previously published research using blockchain in SG systems.

Ref.	Summary	Method/protocol used	Role
[17]	Proposed a certificateless aggregate signature scheme integrating blockchain technology for fog-enabled SGs, improving scalability and reducing latency.	Certificateless aggregate signature scheme with Hyperledger Fabric blockchain.	Achieves authentication, integrity, and privacy while preventing a single point of failure.
[20]	Introduced BPPS, a blockchain-enabled privacy-preserving scheme for secure demand response management using mutual authentication and ECC.	ECC-driven BPPS approach.	Uses mutual authentication, ensures data integrity, and provides secure demand response management.
[21]	Designed an anonymous authentication and key-agreement protocol for SG edge computing using blockchain, supporting revocation and conditional anonymity.	Blockchain-based mutual authentication with dynamic key management.	Enables flexible key management and privacy protection at the edge level.
[22]	Developed a certificate-based access-control scheme for demand response management in SG edge computing, ensuring efficient and secure data exchange.	ECC-based certificate protocol: demand response management authentication scheme (DRMAS).	Provides mutual authentication, fast certificate verification, and resilience to known attacks.
[24]	Investigated physical-layer vulnerabilities in SGs, identifying threats such as jamming, false-data injection, and signal spoofing.	Advanced cryptography and ML.	Secure communication and device-level interaction.
[25]	Proposed a blockchain-based mutual authentication protocol to address security challenges in SG systems.	Blockchain with ECC and dynamic join-and-exit mechanisms.	Improved security and performance through batch verification and Schnorr signatures.
[26]	Proposed a blockchain-based scheme with smart contracts to enhance grid resilience by enabling secure, automated, and auditable energy exchanges.	Blockchain-based scheme for grid resilience.	Maintains trust and data integrity and reduces cyber risks.
[27]	Suggested an SG system using the IoT, blockchain, and edge computing for enhanced functionality and communication.	Integrates IoT and blockchain technology using edge computing.	Maintains data integrity and robust cyber threat resilience.
[34]	Proposed an identity-based designated verifier anonymous aggregate signature (IB-DVAAS) scheme to collect user data securely in SG systems.	Identity-based cryptography with designated verifier and aggregate signatures.	Ensures privacy through aggregate verification and reduces data-center load.
[35]	Introduced a lightweight ECC-based authentication protocol for SG communication with anonymity and mutual authentication.	ECC-based lightweight authentication protocol.	Minimizes computational cost while ensuring secure and anonymous communication.

Tendermint structures, to control efficiency. Meanwhile, smart contracts will autonomously handle demand response, billing, and peer-to-peer (P2P) energy trading.

Blockchain technology can be used to revolutionize SG systems by enhancing transparency and security, building user trust, and efficiently managing grid networks. A detailed analysis of existing work is presented in Table 2. The studies outlined in this section show that blockchain technology can be used to address various security and privacy challenges in SG systems. It can improve storage of energy data, increase trust, and provide real-time pricing. Furthermore, it can automate processes such as billing and transactions, help to maintain grid stability, and provide transparency for renewable-energy certification. The cryptographic protocols involved in blockchain technology can ensure secure data transfer by using ECC-based authentication for user validation, manage the balance between supply and demand, and establish access-control policies in SG systems. Nonetheless, there are still challenges that must be addressed to enhance the efficiency of these systems, such as scalability, latency, energy consumption, conflicts with sustainability goals, and complex integration with existing infrastructure—including IoT devices and legacy systems.

3. Proposed decentralized, blockchain-based SG security framework

Fig. 2 presents our proposed SG architecture based on an edge-enabled, blockchain-integrated system with smart contracts for automation. The decentralized architecture and peer-to-peer (P2P) transactions make the proposed system more trustworthy and reliable than traditional systems. The SG system registers consumers through a trusted authority. Edge nodes perform processing and computation for the consumer, sending the required attributes to the local blockchain via smart contracts. Smart contracts are self-executable programs with a range of use cases [37]; in SGs, smart contracts provide automation, security, and decentralization for energy management. They enable real-time energy trading without intermediaries, lowering operational costs and accelerating settlements. The blockchain is used to record all energy exchanges, ensuring transparency and providing tamper-proof billing. Smart contracts enhance the reliability of the grid by automating tasks such as balancing supply and demand and issuing maintenance notifications. The programmability of these systems supports innovative models such as P2P energy trading and dynamic pricing, promoting efficient and sustainable energy use.

Blockchain operates on a decentralized network that enables P2P transactions and communication. Every node maintains a copy of the blockchain, so if one node fails, the network can continue functioning normally because other nodes will retain all of the required

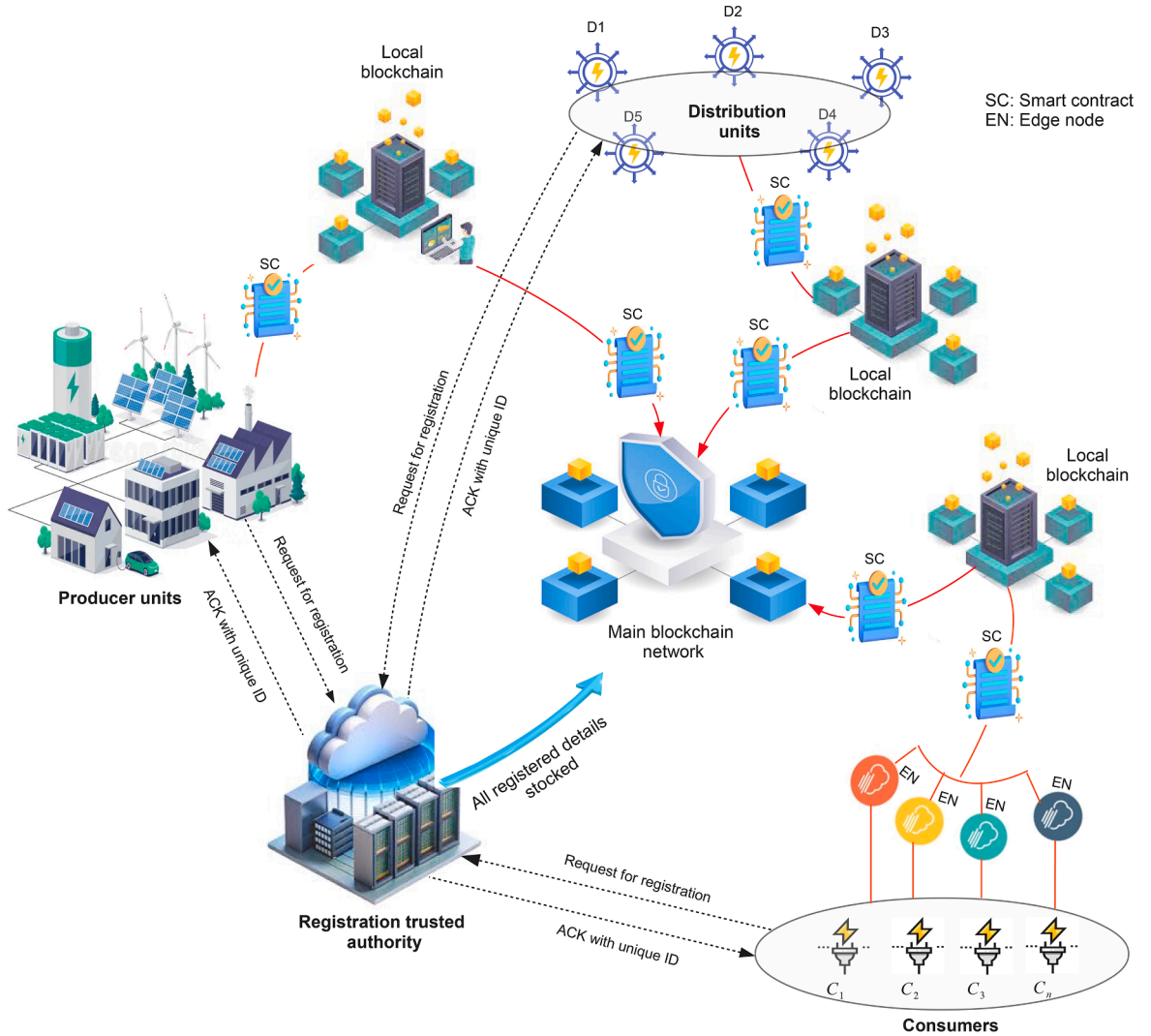


Fig. 2. Diagram of proposed blockchain-based decentralized SG security framework.

information. Unlike centralized networks, in which a single server failure can disrupt the entire system, decentralized blockchain networks do not have a single point of failure.

3.1. System design

This section introduces the design of the proposed system to enhance security and privacy in SG networks. The model comprises four units: consumer units (in which edge computing processes consumer details at the network level), energy-producer units, energy-distribution units, and a trusted registration authority (TRA). Initially, the TRA registers all consumers, producers, and distribution units, assigning each a unique identifier and a key pair. The registration details are added to the main blockchain for future identity verification. Next, the edge nodes collect information from consumers, such as their electricity consumption, usage time, capacity, and behavior. These details are verified through a self-executable smart contract and then added to the local blockchain. At the edge node, the local blockchain stores these data and performs verification to detect any abnormalities. Information from all three types of unit is stored in the local blockchain, and all local blockchains are ultimately connected to the main blockchain network. This layered approach helps to overcome scalability challenges.

In this system, ECC authentication techniques are used to register consumer, producer, and distribution units through a trusted party. [Algorithm 1](#) explains the authentication process for producers and consumers. ECC is used because it offers a secure, lightweight mechanism suitable for distributed networks with resource-constrained smart devices. The details of the ECC mechanism and its execution are provided by Park et al. [20]. The notation used in the proposed scheme is listed in [Table 3](#).

Table 3
Notation used in the proposed framework.

Symbol	Description
Pub_{prod}	Public key of energy-production unit.
Pvt_{prod}	Private key of energy-production unit.
g	Generator of the group used in the cryptographic scheme.
n	Prime modulus.
$H(M)$	SHA-256 hash used for message M .
Sign	Unique digital signature.

Algorithm 1 Authentication of producers and consumers.

```

1: Input:
2:   Producer's message  $M_P$ ,  $\text{Pvt}_i$ ,  $\text{Pub}_i$ .
3:   Consumer's message  $M_C$ ,  $\text{Pvt}_j$ ,  $\text{Pub}_j$ .
4: Output: Entity authentication.
5: procedure SIGNMESSAGE( $M$ ,  $\text{Pvt}$ )
6:    $h \leftarrow H(M)$ 
7:    $\text{Sign} \leftarrow h^{\text{Pvt}} \bmod n$ 
8:   return Sign.
9: end procedure
10: procedure VERIFYSIGNATURE( $M'$ , Sign, Pub)
11:    $h' \leftarrow H(M')$ 
12:   if  $h' = \text{Sign}^{\text{Pub}} \bmod n$  then
13:     Entity authenticated.
14:   else
15:     Authentication failed.
16:   end if
17: end procedure

```

Public-key cryptography: The authentication process is based on public-key infrastructure. Each producer (node) in the SG system possesses a public key Pub_{prod} and a private key Pvt_{prod} . These keys are mathematically related such that $\text{Pub}_{\text{prod}} = g^{\text{Pvt}_{\text{prod}}} \bmod n$, where g is a group generator used by the cryptographic algorithm, Pvt_{prod} is kept secret, and n is a prime modulus.

Digital signature process: Let M represent the message generated by the energy producer. The energy producer authenticates M by signing it. First, the message is hashed using a cryptographic hash function such as SHA-256 to ensure consistent size and to detect any modifications. The hash of the message is denoted by $h_M = H(M)$. Only the hash is signed to improve efficiency. The energy producer signs the message hash using their private key, yielding the digital signature $\text{Sign} = h_M^{\text{Pvt}_{\text{prod}}} \bmod n$.

Verification process by blockchain nodes: Once the energy producer has signed the data and it has been transmitted to the blockchain network, other nodes (validators) must authenticate the energy producer and validate the integrity of the data. This verification involves assessing the authenticity of the digital signature. The verifier, for example a validator node, obtains the public key of the energy producer, denoted as Pub_{prod} . It then computes the hash of the received message; if this hash matches, the message is asserted to be the original message transmitted by the producer: $h_{M'} = H(M')$. To validate the signature, the verifier checks whether the following condition holds: $h_{M'} = \text{Sign}^{\text{Pub}_{\text{prod}}} \bmod n$. If the condition is satisfied, the signature is valid, confirming the message's authenticity and integrity, and verifying that it was signed by the owner of the corresponding private key.

Zero-knowledge proof: Zero-knowledge proof has been implemented to improve privacy by allowing the energy producer to verify their identity without disclosing their private key or any sensitive information. This process is detailed in [Algorithm 2](#).

Blockchain integration using a consensus algorithm: After the message is signed and verified, it is published on the blockchain. Multiple nodes independently verify the producer's identity and the validity of the transaction using a consensus algorithm such as proof of stake, proof of work, or practical Byzantine fault tolerance. Once consensus is achieved, the transaction is recorded in a new block on the blockchain.

Smart contracts play a key role in enhancing data integrity, reducing human error, and automating transactions and validations in smart grid systems. They also strengthen user anonymity by enabling permissionless access, requiring no personal data, and integrating zero-knowledge proofs. In IoT-based smart grids, decentralized authentication is achieved using digital signatures and ECC, with each user or device assigned a unique key pair for secure identity verification, thereby preventing the disclosure of private information. Blockchain technology further supports data security through its immutable structure and consensus mechanisms, which prevent unauthorized alterations. Digital signatures ensure that data is traceable to its sender, reinforcing trust. The use of public key pseudonymity and zero-knowledge proofs boosts user anonymity, allowing claim verification without compromising personal data. Together, blockchain, smart contracts, and ECC address critical security challenges in smart grids, including authentication, data integrity, and user privacy.

Algorithm 2 Zero-knowledge proof.

```

1: procedure ZEROKNOWLEDGEPROOF(Pvt, Pub)
2:   Producer (or consumer) chooses random  $r$ 
3:    $X \leftarrow g^r \bmod n$ 
4:   Validator sends random challenge  $c$ 
5:   Compute  $y \leftarrow r + c \cdot \text{Pvt} \bmod n$ 
6:   if  $g^y = X \cdot \text{Pub}^c \bmod n$  then
7:     Private-key possession verified.
8:   else
9:     Proof failed.
10:  end if
11: end procedure

```

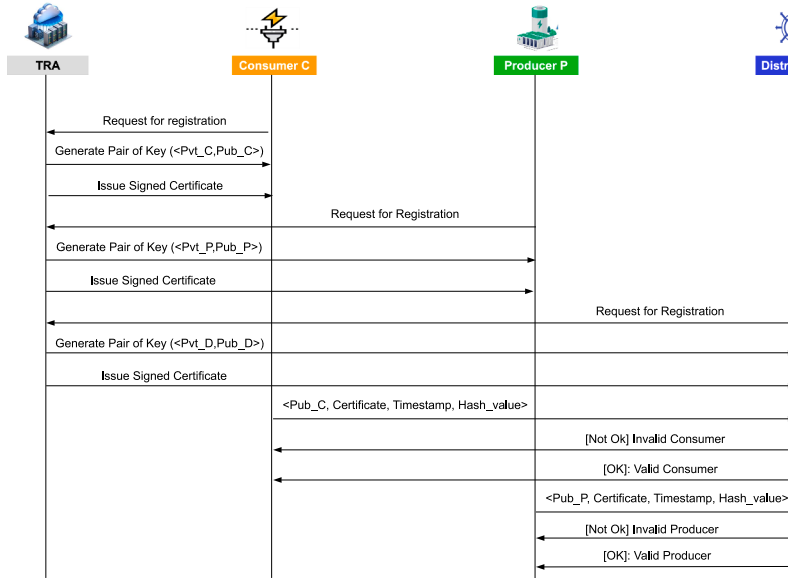


Fig. 3. TRA authentication and verification steps with all three units of the SG.

3.2. Registration and credential generation of users

Each power station (producer or consumer) acts as a feeder, sending transactional data and recording it in its local blockchain network. Each node (hub station) is part of a shared ledger and acts as a peer to perform actions. Each node records power production, losses, and consumption in the local blockchain ledger. A smart-contract-enabled model identifies all operations, detects unauthorized access, and maintains node consensus. Blockchain can be used to store attributes such as carbon emissions, leakage, and losses related to power production, and it can track electricity output down to the kilowatt-hour level. A blockchain network can also incorporate additional sensors to store information on weather conditions and grid infrastructure status, aiding utility maintenance efforts. Blockchain-based smart-business logic can identify surplus energy usage and promote its consumption within local communities before distributing it elsewhere, resulting in significant cost savings.

As shown in Fig. 3, the TRA uses ECC to generate key pairs for users in the SG. We use ECC-based authentication because it supports resource-constrained devices while maintaining sufficient security [20]. The state diagram in Fig. 3 outlines a secure registration and verification protocol involving three entities: consumers (C), producers (P), and distributors (D), each interacting with the TRA. Each entity initiates a registration request and generates a cryptographic key pair: a private key Pvt_X and a public key Pub_X , where $X \in \{C, P, D\}$. The TRA issues a signed digital certificate $\text{Cert}_X = \text{Sign}_{\text{TRA}}(\text{Pub}_X \parallel \text{ID}_X \parallel T_X)$ that binds the entity's identity ID_X and timestamp T_X to its public key.

For verification, the entity submits a tuple $\langle \text{Pub}_X, \text{Cert}_X, T_X, H_X \rangle$, where $H_X = \text{Hash}(\text{Pub}_X \parallel \text{Cert}_X \parallel T_X)$ to ensure data integrity. The TRA verifies the signature using its public key and checks the hash. If valid, the entity is marked as trusted; otherwise, it is marked as untrusted. This process ensures system integrity, mutual authentication, and nonrepudiation using cryptographic techniques such as asymmetric key generation, digital signatures, and hash functions.

Algorithm 3 Smart contract for a SG.

```

1: Struct Consumer
2:   string name;
3:   address consumerAccount;
4:   uint256 amountPaid;
5:   uint256 electrocoins;
6: mapping (bytes32 → Consumer) consumers;
7: Struct Producer
8:   uint256 rate;
9:   string location;
10:  uint256 coinBalance;
11:  uint256 lastActivated;
12:  uint256 lastDuration;
13: mapping (uint → Producer) producers;
14: State variables
15:   address owner;
16:   uint256 numConsumers;
17:   uint256 numProducers;
18:   uint256 coinRate;
19:
20: procedure ENERGYPRODUCER
21:   owner ← msg.sender;
22:   numConsumers ← 0;
23:   numProducers ← 0;
24:   coinRate ← 1018;
25: end procedure
26: procedure REGISTERCONSUMER(_email, _name)
27:   email ← stringToBytes(_email);
28:   if consumers[email].consumerAccount > 0 then
29:     throw;
30:   end if
31:   c ← consumers[email];
32:   c.consumerAccount ← msg.sender;
33:   c.name ← _name;
34:   c.electrocoins ← 0;
35:   numConsumers ← numConsumers + 1;
36: end procedure

```

4. Implementation and evaluation

To conduct an experimental analysis of our proposed technique, we used edge nodes with the following system configuration: Intel Core i7-8700T running at 2.40 GHz with 8 GB of RAM on a 64-bit operating system. The blockchain network was built on Ethereum's open-source platform to establish a decentralized network. Blockchain, with its secure platform and robust security features, is widely used across various applications. Its use of cryptographic hash functions, digital signatures, verification, validation, and mining makes it one of the most secure networks.

In this work, we analyzed the gas utilization of smart contracts deployed on the Ethereum test network. The associated smart contract manages the decentralized application (DApp) logic on the blockchain platform. Unlike cloud applications, DApps operate in a decentralized manner without a single authority controlling the application's infrastructure. DApps can also provide user-friendly interfaces to smart contracts, allowing users to submit transactions to the contract via a web interface. To access a DApp, users simply enter the URL or server address into a browser.

We implemented the smart contract using Solidity, a programming language similar to Java. Each edge device runs a controller service on the IoT device that polls the SG smart contract described in [Algorithm 3](#) at regular intervals, checking the status of the production source. Each production unit has a unique associated identifier (ID).

This section comprehensively explains the implementation of the proposed method through smart contracts. The experimental setup includes four computer systems acting as power generators or distributors, and two lightweight devices to connect the equipment. We built the blockchain network using the Ethereum platform and wrote the proposed system's smart contracts in Solidity. These smart contracts were developed and tested in the Remix integrated development environment (IDE) before deployment on the blockchain platform.

Algorithm 4 shows the Solidity implementation of the edge-node smart contract, which specifically handles consumer–producer registration. We wrote the smart contract using the Remix Solidity environment and deployed it on the Ethereum network. The Ethereum network supports node creation, with each node acting as either a consumer or producer in the SG system.

Algorithm 4 Smart contract for consumer data management in a SG.

```

1: contract Edgenode
2: struct Consumer
3:     uint256 id;
4:     uint256 consumedUnits;
5:     uint256 paidAmount;
6:     uint256 lastUpdated;
7: mapping(address => Consumer) public consumers;
8: event ConsumerUpdate(
9:     address indexed consumerAddr,
10:    uint256 consumedUnits,
11:    uint256 paidAmount,
12:    uint256 timestamp
13: ); // Modifier ensures that only an authorized edge node can update smart-meter data to address public owner;
14: modifier onlyOwner() {
15:     require(msg.sender == owner);
16: }
17: Constructor() {
18:     owner = msg.sender;
19: }
20: function updateConsumerData(address _consumerAddr, uint256 _consumedUnits, uint256 _paidAmount) public onlyOwner {
21:     require(_consumerAddr != address(0),
22:         "Invalid consumer address");
23:     Consumer storage consumer = consumers[_consumerAddr];
24:     consumer.id = uint256(uint160(_consumerAddr));
25:     consumer.consumedUnits = _consumedUnits;
26:     consumer.paidAmount = _paidAmount;
27:     consumer.lastUpdated = block.timestamp;
28:     emit ConsumerUpdated(_consumerAddr, _consumedUnits, _paidAmount, block.timestamp);
29: }
30: function getConsumerData(_consumerAddr)
31: public view
32: returns (uint256, uint256, uint256, uint256)
33: { Consumer memory consumer = consumers[_consumerAddr];
34: return (
35:     consumer.id,
36:     consumer.consumedUnits,
37:     consumer.paidAmount,
38:     consumer.lastUpdated );
39: }
40: }

```

4.1. Informal security analysis

In this subsection, we present a comparative analysis of different security challenges based on the issues listed in [Table 1](#).

- **Authentication:** A decentralized SG network connects users such as consumers, producers, and distributors to the local network. Here we employ an ECC-based authentication scheme to authenticate all users via a TRA. After registration, user identities can be verified. Even if the trusted unit is compromised by attackers, user identities can still be confirmed from the immutable digital ledger stored on the network.
- **User anonymity:** In an SG network, users generate large amounts of data that must be protected. Improper handling of this data can expose sensitive information to attacks. In this proposed blockchain-based approach, user anonymity is ensured through zero-knowledge proofs, allowing users to validate transactions and data without revealing their identity. Additionally, edge nodes manage local pseudonym generation and mapping. User data is anonymized before transmission to the blockchain, ensuring that users remain untraceable.

Table 4
Comparison of computation cost (ms).

Scheme	Total computation cost (ms)
[17]	1.809
[20]	8.929
[21]	80.15
[22]	20.11
[34]	10.74
[35]	11.17
Proposed	3.150

- **Edge:** We use low-end devices to process information at the network edge, enabling rapid validation of user and transaction identities. This reduces the load on the main network and supports real-time scenarios by integration with self-executable smart contracts.
- **Distributed data storage:** Centralized storage solutions are vulnerable to single points of failure and scalability issues. Here, we advocate for decentralized blockchain storage for local and global networks, mitigating the risk of data loss and improving scalability. Edge nodes transmit all valid transactions as blocks into the local blockchain network and function as partial nodes, minimizing latency and bandwidth demands.
- **Privacy preservation:** Privacy challenges arise when data is communicated or transferred in plain text. Our proposed solution addresses these challenges using a private blockchain that authenticates all devices and users on the network. Only authorized users and smart devices can communicate or transmit information, and all registered users can participate in communications or verify and validate transactions within the network.
- **Data integrity:** Manipulating SG data, including energy-consumption metrics, can cause financial losses or operational disruption. The immutability of blockchain data guarantees that once it is recorded, it cannot be altered without consensus. Edge nodes act as a primary safeguard by locally verifying data before submitting it to the blockchain, reducing complexity in the blockchain network.

4.2. Performance evaluation

Blockchain technology is used in SG systems to improve the security, transparency, and efficiency of energy management. Its performance is evaluated using three key cost metrics: computation cost, communication cost, and smart-contract execution cost. Computation cost measures the resources required for energy transactions, while communication cost assesses data exchange among grid nodes. Smart-contract execution cost is critical, encompassing both one-time deployment costs and ongoing costs for regular interactions.

4.2.1. Computation cost

In this part, we evaluate the computational performance of the proposed scheme against current schemes. The computational cost of cryptographic operations is the sum of the time required for signature generation and verification. As shown in Table 4, the computation cost is improved by 64.7 % compared to Ref. [20], 96.07 % compared to Ref. [21], 84.34 % compared to Ref. [22], 70.65 % compared to Ref. [34], and 71.80 % compared to Ref. [35]. Although Ref. [17] outperforms our proposed scheme in terms of computation cost, our scheme performs better in terms of communication cost and offers improved security through a blockchain network. Overall, the proposed scheme shows improved performance compared to several previous works. Employing a lightweight ECC algorithm alongside a single-round mutual authentication protocol reduces computational cost and enhances efficiency in the verification process for smart-grid applications. These applications benefit from tamper-evident, secure verification while simultaneously reducing processing time.

4.2.2. Communication cost

In this section, we examine the communication overhead associated with the proposed blockchain-enabled scheme and compare it with the latest state-of-the-art schemes designed for IoT-based smart grids. Communication overhead refers to the additional size of security parameters, such as random numbers, hash outputs, timestamps, and identity credentials, that must be transmitted alongside the message, impacting both bandwidth and communication latency. In this work, the total combined size of the random number, hash function, and identity is taken as 160 bits, with a timestamp of 32 bits and an elliptic-curve point of 320 bits [20]. In our proposed system, users register with the trusted authority and receive a unique ID, which they use to communicate with the edge node, resulting in a total communication cost of 992 bits. As shown in Fig. 4, the proposed scheme requires fewer bits for communication. It illustrates the comparative analysis of communication costs between the proposed scheme and prior schemes. The communication costs of existing SG systems are: 1152 bits [17]; 1376 bits [20]; 3456 bits [21]; 1344 bits [22]; 1120 bits [34]; and 576 bits [35]. The proposed scheme's communication cost is 13.89 % less than that of Ref. [17], 27.91 % less than Ref. [20], 71.30 % less than Ref. [21], 26.19 % less than Ref. [22], and 11.43 % less than Ref. [34]. Although the communication cost is higher than in Ref. [35], the proposed scheme delivers lower computation cost and blockchain-backed auditability, which we argue is preferable for tamper-evident logging in SG settings. Overall, the proposed scheme improves upon existing systems while providing enhanced security using blockchain technology. To sum up, the proposed scheme showed a notable decrease in communication overhead compared to most

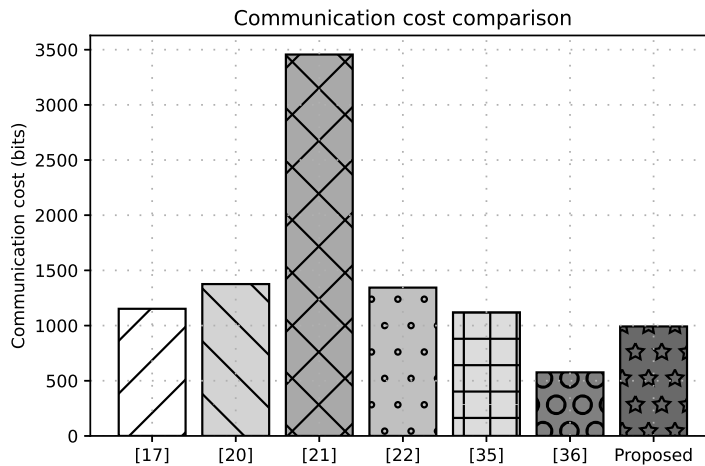


Fig. 4. Comparison of communication costs with existing work.

Table 5
Smart-contract transaction cost (fee).

Scheme	Gas used	Ether cost	USD
Deployment:			
Ref. [20]	90,551	0.001448816	6.30
Proposed	49,774	0.001493220	5.64
Writing:			
Ref. [20]	43,528	0.0000696448	3.03
Proposed	25,527	0.000765810	2.89

current research efforts. Moreover, it offers enhanced blockchain-backed accountability and reduced computational load. Together, these characteristics constitute an achievement where security and performance coexist, positioning it as a foremost framework for next-generation SG applications.

4.3. Smart-contract execution

In this section, we describe the simulation of the proposed scheme using the Remix IDE to test its feasibility and efficiency. Remix IDE is a native development environment for “Web 3.0” applications; it is available as an open-source web and desktop application for the Ethereum network. The smart contract in Algorithm 4 is executed by the edge node. For this, we used the Remix Solidity compiler version 0.8.28 + commit.40a35a09.

With the results depicted in Table 5, we can evaluate the transaction fees associated with deploying and conducting write operations based on this simulation. Transaction gas costs are incurred by operations performed on the Ethereum testnet. At the time of analysis, the price of ETH was 3,774 USD, significantly higher than average. This rise in ETH price also increased the average gas cost, as gas prices fluctuate with market conditions and network demand.

This analysis was conducted using an average gas price of 30 Gwei. As shown in Table 5, the cost of deploying a smart contract is 5.64 USD, compared to the 6.30 USD reported in Ref. [20]. Similarly, the cost of writing data is 2.89 USD per transaction, which is slightly higher than that in Ref. [20]. The blockchain’s data-writing cost is typically approximately 2.89 USD for a single operation. The obtained results demonstrate low communication and computation costs, as well as efficient execution of smart contracts on the Ethereum decentralized network. The results indicate that the associated gas fees are minimal when compared to the potential losses caused by attacks on the smart grid (SG). These findings suggest that the proposed method is both cost-effective and practical for real-time SG security, given that the smart-contract deployment occurs only once. Furthermore, the reduced gas utilization highlights the lightweight and efficient design of the proposed smart contract, minimizing redundant on-chain operations and ensuring cost-effective execution. This efficiency ensures scalability when integrated with multiple edge nodes and supports sustainable, low-cost blockchain operations in long-term smart-grid deployments.

5. Conclusion

Blockchain technology offers many advantages, including reduced processing time, increased reliability, and a decentralized, secure platform. Numerous applications have been developed using blockchain to create a trustless environment in which every node participates in decision-making. Blockchain technology is tamper-proof, and the digital ledger it stores is immutable. The approach outlined in this paper uses smart grid (SG) infrastructure as a case study, and we propose a decentralized architecture using

blockchain technology. The paper also discusses the security issues and challenges inherent in traditional SG systems. Ethereum, an open network, is used to deploy smart contracts to implement the proposed architecture. With blockchain and smart contracts, intelligent grid systems can achieve transparency, trust, data integrity, and security.

The proposed scheme has a computational cost of 3.150 ms and a communication overhead of 992 bits, providing improvements over various existing methods. Additionally, the expense for deploying the smart contract is 5.64 USD and the cost for data writing is 2.89 USD, both more economical than comparable approaches. The proposed scheme can be used to ensure resilience against single points of failure by addressing critical aspects such as data integrity, user anonymity, distributed storage, trust establishment, and a decentralized architecture.

Although the proposed decentralized framework for SG deployments offers enhanced security and efficiency, challenges such as network congestion, latency, and resource consumption may arise. Integration with legacy systems and ensuring interoperability between devices and platforms may require custom middleware or extensive retrofitting. Additionally, cryptographic operations can pose challenges for resource-constrained edge devices, requiring robust solutions and widespread adoption in complex SG environments. In the future, integrating AI into the proposed framework could enable intelligent threat detection, adaptive energy optimization, autonomous grid management, and improved decision-making within decentralized, blockchain-secured smart grids.

CRediT authorship contribution statement

Bhabendu Kumar Mohanta: Conceptualization, Methodology, Investigation, Validation, Formal analysis, Data curation, Writing - original draft, Writing - review & editing; **Ali Ismail Awad:** Writing - review & editing, Writing - original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization; **Tarek Elsaka:** Writing - review & editing, Writing - original draft, Resources, Data curation, Conceptualization; **Hamza Kheddar:** Writing - review & editing, Writing - original draft, Visualization, Data curation, Conceptualization; **Ezedin Baraka:** Writing - review & editing, Writing - original draft, Methodology, Conceptualization.

Data availability

No datasets were used for the research described in this article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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